

1	(a)	Upthrust is the weight of the oil displaced by cylinder X. Cylinder X is fully submerged in oil at position X. Hence volume of oil displaced is equal to the volume of the cylinder. Upthrust = (density of oil) $\times$ (volume of cylinder) $\times$ 9.81 $U = 940 \times 0.018 \times 1.2 \times 9.81$ $U = 200 \text{ N}$	2
	(b)	(i) Since velocity is constant throughout the lifting process, net force on cylinder X is zero. Tension + Upthrust = Weight of cylinder $T = 560 - 200$ $T = 360 \text{ N}$	2
		(ii) Power output of motor = $T v$ $= 360 \times 0.020$ $= 7.2 \text{ W}$	2
	(c)	As the cylinder is raised the <u>volume of the liquid displaced decreases</u>. Hence, <u>upthrust acting on the cylinder decreases [B1]</u>. With the <u>weight of the cylinder unchanged</u>, the <u>tension in the wire increases [B1]</u>. Therefore, the power output of the motor increases.	2
2	(a)	FBD at the top	